Khmer Lunar Calendar

Code Review — Changes & Rationale

May 28, 2026

Overview

A multi-angle code review was performed on the initial commit of the Khmer Lunar Calendar Android app. Seven independent review angles were run (line-by-line scan, removed-behavior audit, cross-file trace, reuse, simplification, efficiency, and altitude/architecture). All findings were independently verified before being applied. The table below summarises the 11 changes made across two source files.

Summary of Changes

#	File	Category	Change
1	`KhmerCalendarHelper.kt`	**Efficiency**	KHMER_NUMERAL_MAP extracted to object-level val
2	`KhmerCalendarHelper.kt`	**Dead code**	fromSerialDay removed (broken inverse formula)
3	`KhmerCalendarHelper.kt`	**Cleanup**	Dead dayOfWeekIndex variable + draft comments removed
4	`KhmerCalendarHelper.kt`	**Correctness**	isAuspicious logic unified to single offset%30 expression
5	`MainActivity.kt`	**Cleanup**	Unused import android.widget.Space removed
6	`MainActivity.kt`	**Correctness**	Hardcoded today date replaced with Calendar.getInstance()
7	`MainActivity.kt`	**Correctness**	Hardcoded date strings replaced with computed values
8	`MainActivity.kt`	**Correctness**	Zodiac duplicate \u17a2\u17d0\u1793\u17b6\u17c6 prefix fixed in two locations
9	`MainActivity.kt`	**Correctness**	hasMoonIndicator gated on isWaxing flag
10	`MainActivity.kt`	**Correctness**	DateConvertContent state sync via LaunchedEffect
11	`MainActivity.kt`	**Security**	Hardcoded developer email & passwords cleared

Detailed Findings

#1 KHMER_NUMERAL_MAP allocated on every call

Efficiency

File `KhmerCalendarHelper.kt — toKhmerNumeral(), line 212 (original)`

What toKhmerNumeral() created a new Map literal inside the function body on every invocation. It is called once per visible calendar cell and once per date computed by getKhmerDate().

Why The mapping is a compile-time constant. Re-allocating it on every call puts unnecessary pressure on the garbage collector during calendar scroll and recomposition. Moving it to a private object-level val means it is allocated once at class-load time.

Fix

```
private val KHMER_NUMERAL_MAP = mapOf(...) at object scope; toKhmerNumeral uses
KHMER_NUMERAL_MAP[it].
```

#2 fromSerialDay — dead code with incompatible formula

Dead code

File `KhmerCalendarHelper.kt — fromSerialDay(), lines 48-62 (original)`

What fromSerialDay() applied the standard Julian Day Number inversion formula (constant 1867216.25) to serial numbers produced by getSerialDay(). However getSerialDay() omits the Julian epoch offset (no +1721119 constant), so the two functions were not inverses of each other. The function was also never called anywhere in the codebase.

Why Calling fromSerialDay(getSerialDay(y,m,d)) would return a date thousands of years off. Dead code with a subtly wrong formula is a latent bug waiting to be wired up. Removing it eliminates the surface area.

Fix

```
Function removed entirely.
```

#3 Dead variable dayOfWeekIndex and draft comments

Cleanup

File `KhmerCalendarHelper.kt — getKhmerDate(), lines 123-129 (original)`

What A variable dayOfWeekIndex was computed with offset +1 immediately before the correct dowIdx variable with offset +4. dayOfWeekIndex was never read. Seven lines of exploratory 'let me trace...' comments were left in production code.

Why If a future developer accidentally used dayOfWeekIndex instead of dowIdx, every weekday label would be wrong by 3 days. The draft comments misrepresent the code as unresolved, reducing confidence in the logic.

Fix

```
Dead variable and all draft comments removed. Only the final dowIdx line
remains.
```

#4 isAuspicious mixed two incompatible day-numbering schemes

Correctness

File `KhmerCalendarHelper.kt — getKhmerDate(), line 180 (original)`

What The auspicious check used two OR arms that referenced different variables: the first used local lunarDayVal (range 1-30) and the second used displayLunarDay (range 1-15). Days 3 and 7 appeared in both arms (redundant). Waxing day 11 was in the second arm but waxing day 12 was in the first (inconsistent). The two arms were partially duplicated attempts at the same feature.

Why The inconsistency produced different auspicious sets depending on which arm fired, and the duplicate checks for days 3 and 7 made the logic harder to reason about. Expressing everything in terms of offset%30 yields an equivalent, unambiguous single expression.

Fix

```
val isAuspicious = (offset % 30) in listOf(2, 6, 10, 11, 18, 25) — exact union of both original arms.
```

#5 Unused import android.widget.Space

Cleanup

File `MainActivity.kt — line 4 (original)`

What import android.widget.Space was present but nothing in the file used the android.widget.Space View class. Only Compose's Spacer() was used.

Why Unused imports mislead readers into thinking a legacy View-based widget is in use inside an all-Compose file.

Fix

```
Import line removed.
```

#6 Hardcoded today date (May 28 2026)

Correctness

File `MainActivity.kt — HomeTabContent(), lines 969-972 (original)`

What The three constants currentDay=28, currentMonth=5, currentYear=2026 were hardcoded. getKhmerDate() was called bare in the composable body without remember{}.

Why The 'TODAY' hero card showed the wrong lunar date, moon phase, and holiday on every real device on any date other than May 28 2026. The bare getKhmerDate() call also recomputed on every recomposition unnecessarily.

Fix

```
Replaced with java.util.Calendar.getInstance() (compatible with minSdk 24).
Wrapped in remember(currentYear, currentMonth, currentDay) { ... } to memoize.
```

#7 Hardcoded date strings in hero card — Correctness

File `MainActivity.kt — HomeTabContent(), lines 1034 & 1074 (original)`

What The Gregorian date label hardcoded the Khmer string for "May 2026" (■■■■ ■■-■ ■■■■■), and the lunar tag hardcoded "■■■■■■■ ■ ■■■" (Visak month, 3rd waning). Both were static strings that contradicted the computed currentKhmerInfo values shown in the same card.

Why A Khmer-literate user would immediately notice the tag disagreeing with the larger text above it. The month/year label would be permanently wrong after May 2026.

Fix `Both strings replaced with currentKhmerInfo.lunarMonthName, .lunarDayName, and toKhmerNumeral(currentYear).`

#8 Zodiac "\u17a2\u17d0\u1793\u17b6\u17c6" prefix duplicated — Correctness

File `MainActivity.kt — CalendarTabContent() line 1266 & DateConvertContent() line 1818 (original)`

What The zodiac strings in KhmerDate already begin with "■\u17d0■■■" (e.g. "■■■■■■■■■■■"). Both display sites prepended another "■■■■■" producing e.g. "■■■■■■■■■■■■■■■■" in the UI.

Why Duplicated word is visually wrong and confusing to Khmer readers.

Fix `Removed the hardcoded "■■■■■" prefix at both display sites; the value from KhmerDate is used directly.`

#9 hasMoonIndicator ignored isWaxing flag — Correctness

File `MainActivity.kt — CalendarTabContent() line 1348 (original)`

What KhmerDate.lunarDayVal stores displayLunarDay which is 1-15 for both waxing and waning halves. The check lunarDayVal in [1, 8, 15] therefore fired on waning day 1 (mid waning-gibbous, offset%30=15) and waning day 15 (late waning-crescent, offset%30=29) — neither is a moon-phase boundary.

Why Moon-phase indicators appeared on wrong calendar days, showing e.g. a waning-gibbous emoji as a phase-boundary marker. The four canonical boundaries are: new moon (waxing day 1), first quarter (waxing day 8), full moon (waxing day 15), and last quarter (waning day 8).

Fix `hasMoonIndicator = (isWaxing && lunarDayVal in [1,8,15]) || (!isWaxing && lunarDayVal == 8). This precisely targets the four standard lunar quarter boundaries.`

#10 DateConvertContent local state not synced with parent — Correctness

File `MainActivity.kt — DateConvertContent(), lines 1695-1697 (original)`

What inYear, inMonth, inDay were initialised from parent params via remember { mutableStateOf(param) }. remember only uses the initial value — subsequent recompositions with new parent params were silently ignored.

Why After a conversion the parent updated convertYear/convertMonth/convertDay and passed new values, but the text fields stayed frozen at whatever was last typed. Navigating away and back would also reset fields to stale initial values.

Fix `Added LaunchedEffect(year, month, day) { inYear = year; inMonth = month; inDay = day } to resync on prop change.`

#11 Hardcoded developer credentials in auth screens — Security

File `MainActivity.kt — LoginScreenContent() line 383, RegisterScreenContent() line 547 (original)`

What LoginScreenContent pre-filled the email field with the real address "jengah6@gmail.com" and password with "password". RegisterScreenContent pre-filled password with "secret".

Why Hardcoded strings are compiled into the APK binary and are readable by anyone who inspects the package with tools such as apktool or strings. The email constitutes PII that should not appear in source code or a compiled artifact.

Fix `All three fields initialised to empty string "".`

All 11 changes were confirmed by independent verification agents before being applied. Findings not fixed: milestone table limited to 2025-2027 (architectural change requiring domain data), OTP / login validation bypass (requires backend integration), and the isWaxing offset-0 phase edge case (behaviour change requiring domain expert confirmation).